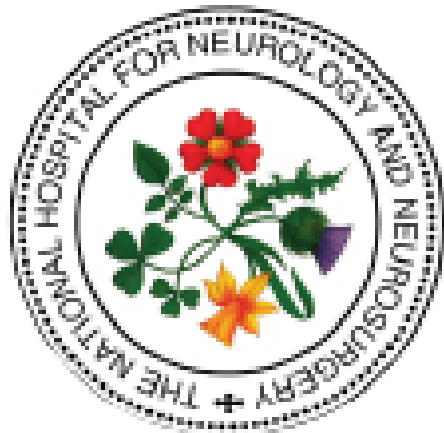




The National Hospital for Neurology and Neurosurgery Neuroscience Critical Care Unit (NCCU)

EXTERNAL VENTRICULAR DRAIN (EVD) INSERTION and MONITORING and CHALLENGING

PRE-OPERATIVE

WARD

- If EVD insertion is a planned procedure, wash hair with hibiscrub on day of operation (*apply neat and lather on scalp*)
- Hair should be thoroughly dried
- MRSA screening and suppression

MRSA status KNOWN +ve

- Suppression therapy
- Wash skin around operation site with chlorhexidine
- Teicoplanin 400mg IV + gentamicin 1.5mg/kg

MRSA status UNKNOWN

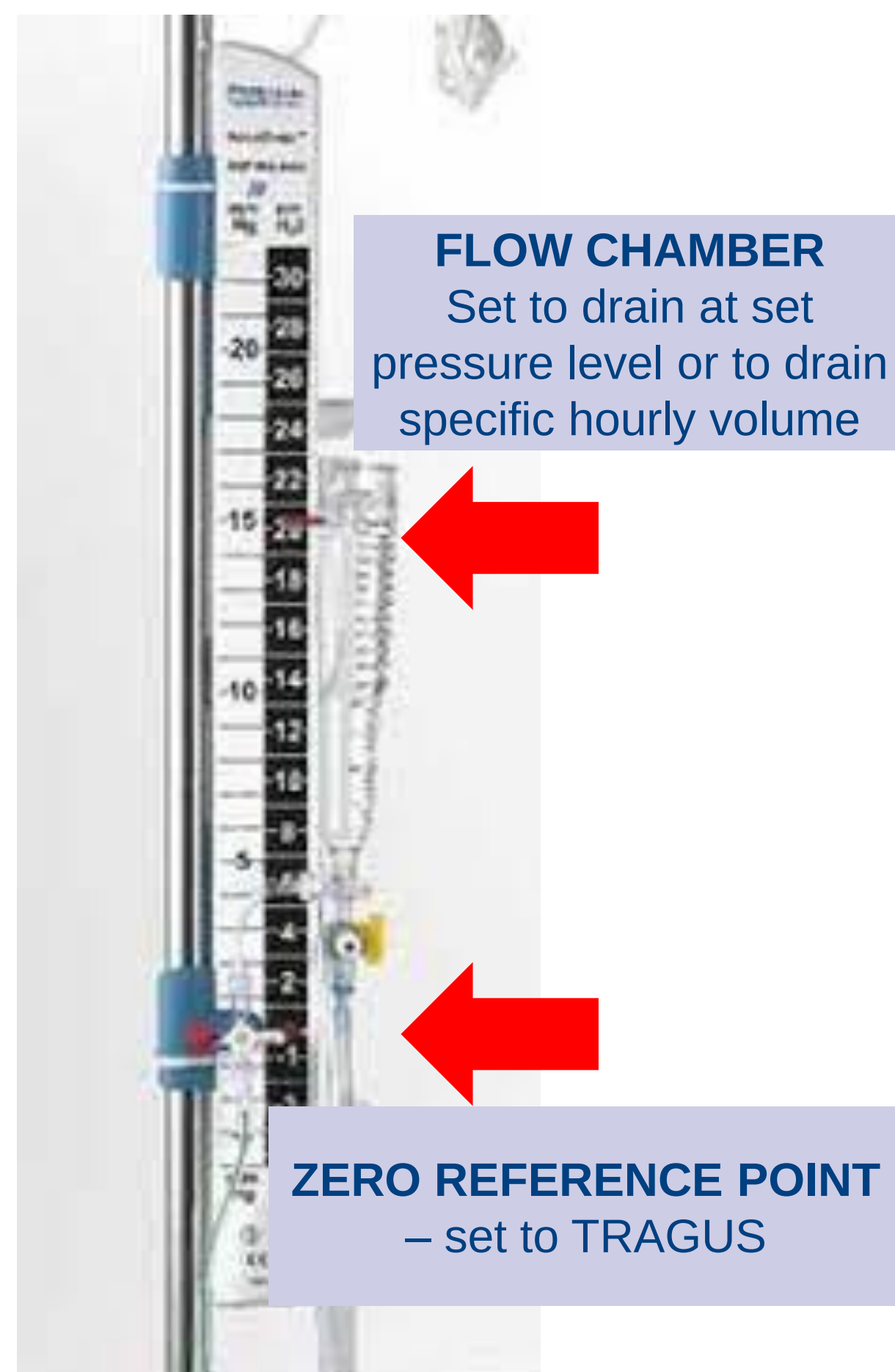
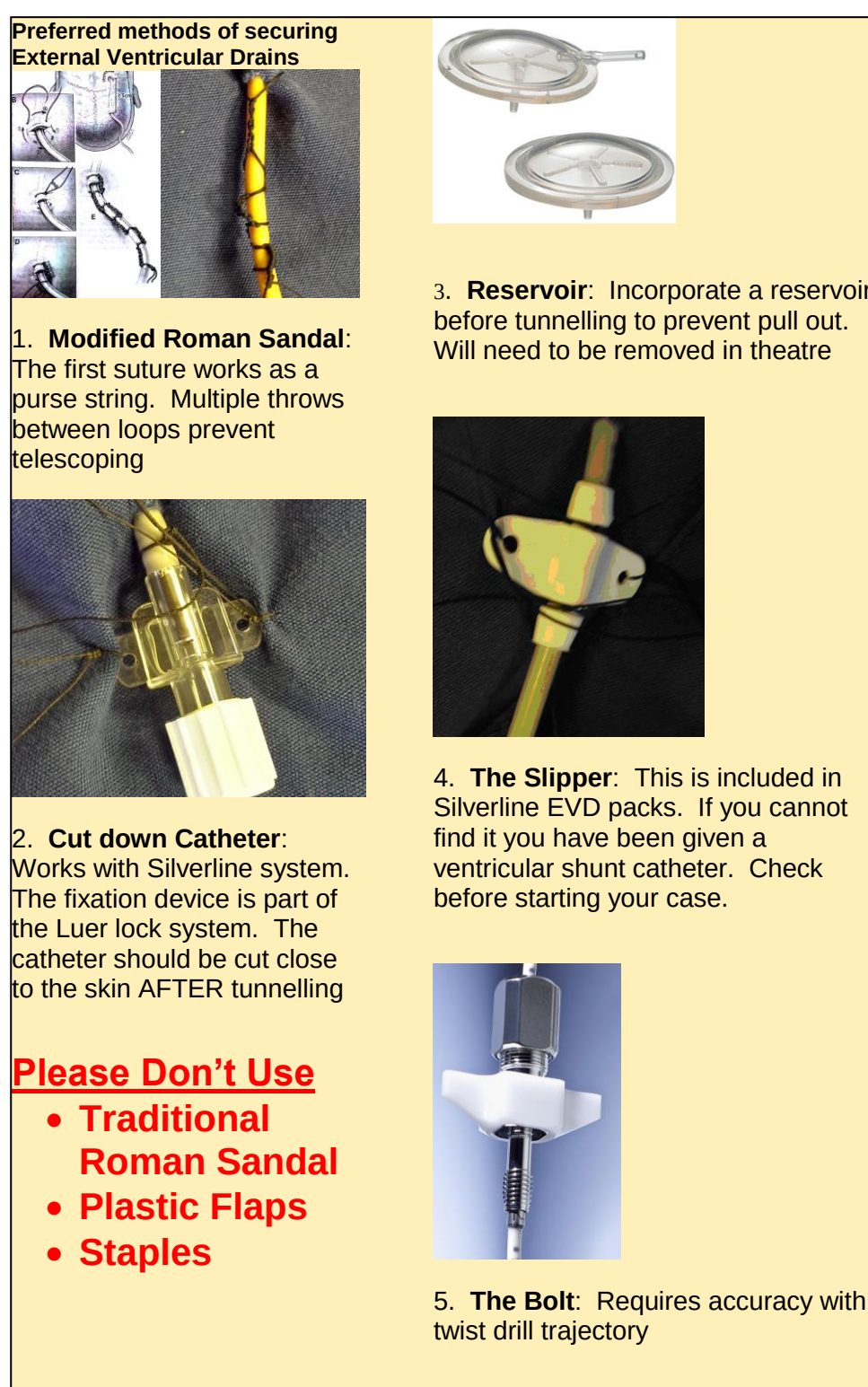
- Take MRSA screen
- Pre-op apply mupirocin to inside of nose
- Wash skin around operation site with chlorhexidine
- Add Teicoplanin 400mg IV to conventional surgical prophylaxis
- Continue MRSA suppression post-op until screen reported

INTRA-OPERATIVE

THEATRE

- Hair clipping as near to time of surgery as possible
- Patient transferred to operating table unless extreme emergency
- Surgical antibiotic prophylaxis
- Take CSF samples on EVD catheter insertion (*2 plain tubes with 5mL CSF in each + 1 sugar tube*)
- Connect EVD drainage system while patient on operating table under sterile conditions
- Dress wound with sterile absorbent adhesive dressing
- Drainage clamps usually opened at final destination with instructions on drainage **height** or **volume per hour**

PREFERRED METHODS of SECURING EVD



POST-OPERATIVE

1. ZEROING and POSITIONING EVD

- Neurosurgical post-op instructions
- Set pressure level or Volume of CSF** drainage each hour

*EVD is a **gravity** drain with a **zero reference point** therefore position determines amount of CSF drained*

- Nursing staff caring for patient with EVD must have been assessed as competent

Zeroing EVD

- Estimating level of lateral ventricle
- External reference point: TRAGUS** - set zero on manometer to this point

Setting pressure level of EVD

- Set pressure level on drip chamber to required height

2. RECORDING CSF DRAINAGE

- CSF drainage should be monitored and recorded hourly
- Note colour and consistency of CSF
- If CSF drainage low (**<5mL/h**) check drain not set too high
- If CSF drainage high (**>20mL/h**) check drain not set too low
- If consistently draining >20mL/h d/w neurosurgical team

3. MOVING / TRANSPORTING PATIENT with EVD

- EVD kept upright and open **if possible**, to maintain drainage
- If **not** possible to transport upright, flow chamber should be emptied and drain clamped while patient moved (*to maintain integrity of antimicrobial filter*)
- When repositioning patient, EVD should be clamped and then re-zeroed after procedure
- Clamp EVD during physiotherapy (*provided no longer than 20 minutes*)
- Patient with EVD can sit up or out of bed provided drain re-zeroed

4. CHANGING DRAINAGE BAG

- Drainage bag **only changed when full** to minimise interruption of closed system (ANTT procedure)

5. REMOVAL of EVD

- Removed by neurosurgeon (ANTT procedure)
- Do **not** send catheter tip for culture unless specifically directed by microbiology consultant (*tip can be contaminated with skin flora on removal*)
- After removal, EVD site should be checked for CSF leak, protected with occlusive dressing / observed for signs of infection

EVD CHALLENGE

ASSESS EVD DAILY

- Stable neurology
- No planned interventions (coiling, extubation) that may cause neurology change
- CSF has no fresh blood or clots
- No evidence of infection / microbiology concur

08:00 CLAMP EVD

Monitor Neurology

CT HEAD at 15:00 or in 24 HOURS

CT stealth with EVD **clamped**

Perform CT (stealth) earlier if

- Neurological deterioration or
- CSF bypasses drain

Failed Challenge

- Neurological deterioration
- CSF bypasses drain
- If safe, scan prior to reopening EVD

No Hydrocephalus on CT

Hydrocephalus on CT

Clinical decision

- Re-open EVD and
- Re-challenge in 48h **or**
 - Proceed to shunt if no evidence of infection

Successful Challenge

- Keep EVD clamped until removal next morning
- Measure ICP and take CSF sample prior to removal